

Cellular Expert

7. MW Equipment

1. Objective

This tutorial will show you how to add new MW links, manage and do predictions.

At the end of the exercise you will be able to:

- Create and import MW equipment.
- Create MW links in the project.
- Do MW Predictions.

2. Initial data

Prepared project with:

- Geodata.
- Equipment and models.

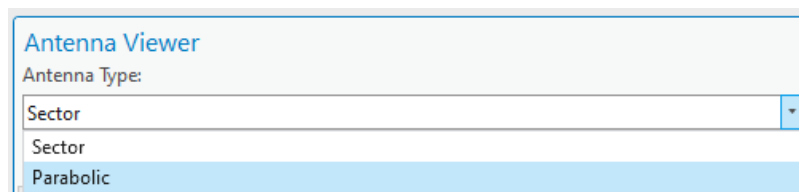
3. Manage MW equipment

Navigate to C:\CE_Course\MW_Equipment\Project and run Project.aprx file to open the prepared project for RL Introduction exercise.

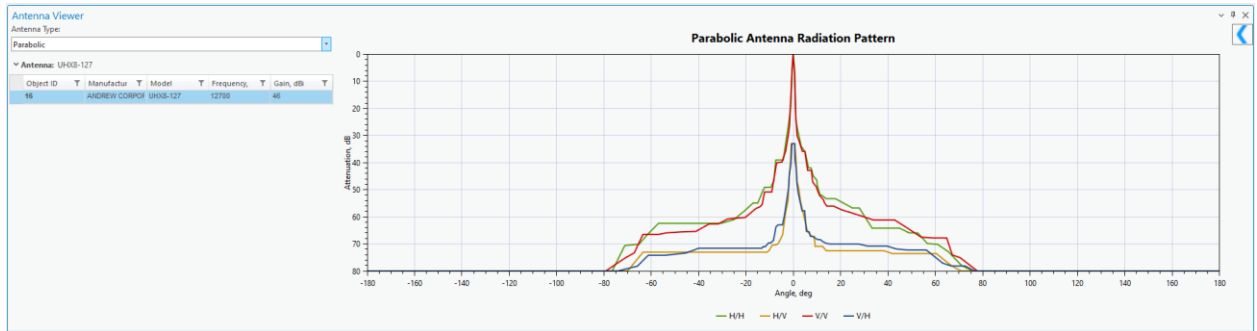
Microwave links involve additional equipment settings compared to point-to-area predictions. These settings include frequency plans, radio equipment, and parabolic antennas, which are used for predicting power budget, interference, or availability. In this section, we will cover how to create and, if necessary, import this data into the project.

3.1 Parabolic antennas

Open Antenna Viewer in CE RLP tab, and change antenna type from Sector to Parabolic.



All parabolic antennas will be displayed.



Close Antenna Viewer tool. To import a new parabolic antenna, open Import/Export Antenna Files tool. Parabolic antenna import supports:

- Andrew format:

```
REVNUM: ,NSMA WG16.99.050
REVDAT: ,19990520
ANTMAN: ,ANDREW CORPORATION
MODNUM: ,KP8F-335
PATNUM: ,2119
FEDORN: ,NA
DESCR1: ,
DESCR2: ,
DESCR3: ,
DESCR4: ,
DESCR5: ,
DTDATA: ,19960617
LOWFRQ: ,335
HGHFRQ: ,365
GUNIT: ,DBI/DBR
LWGAIN: ,15.6
MDGAIN: ,15.7
HGGAIN: ,15.8
AZWIDT: ,19.60
ELWIDT: ,21.40
ATVSWR: ,1.30
FRTOBA: ,16.0
ELTILT: ,0
PATTYP: ,ENVELOPE
```

- NSMA format:

```

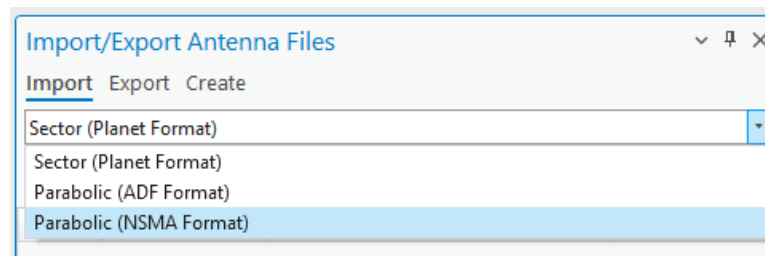
Tutorial Radiolinks
Antenna 10GHz

Cellular Expert Demo Data

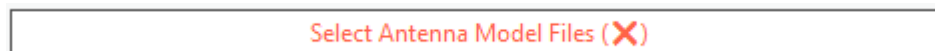
10000-10000 MHz
38.0 dBi
0.0 DEGREES
HH      43
-180.00 -64.00
-100.00 -64.00
-90.00  -58.00
-80.00  -51.50
-67.70  -43.70
-50.00  -39.00
-45.00  -38.50
-35.00  -36.00

```

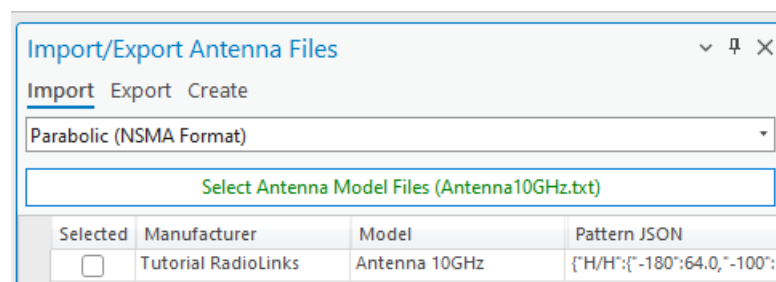
Select Parabolic (NSMA Format).



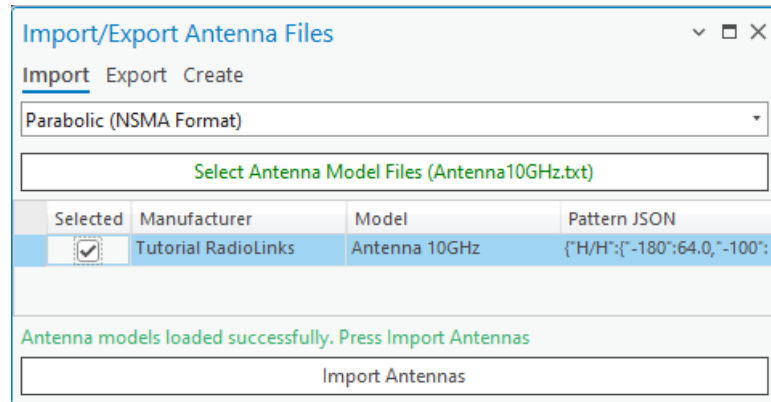
Click on Select Antenna Model Files



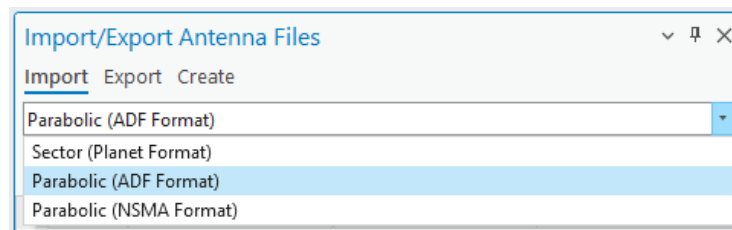
Browse to C:\CE_Course\RL_Prediction\Equipment\Antennas, select Antenna10GHz.txt file and press OK button. It will appear in Import/Export Antenna Files dialog.



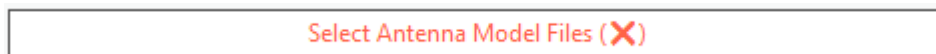
Select Check box near the antenna and press Import Antennas button.



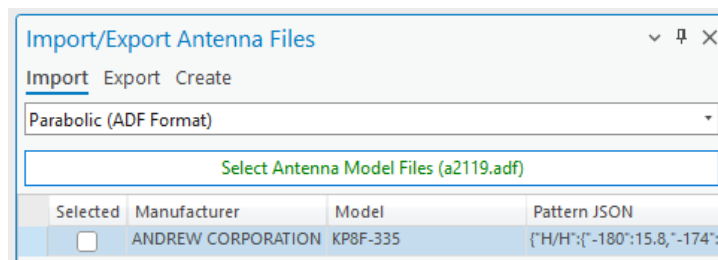
Antenna will be imported successfully. Now choose Andrew format.



Click on Select Antenna Model Files

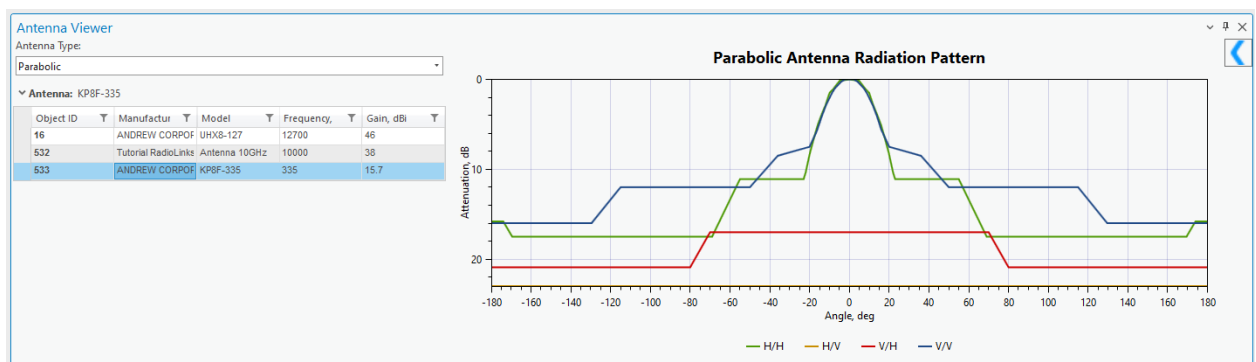
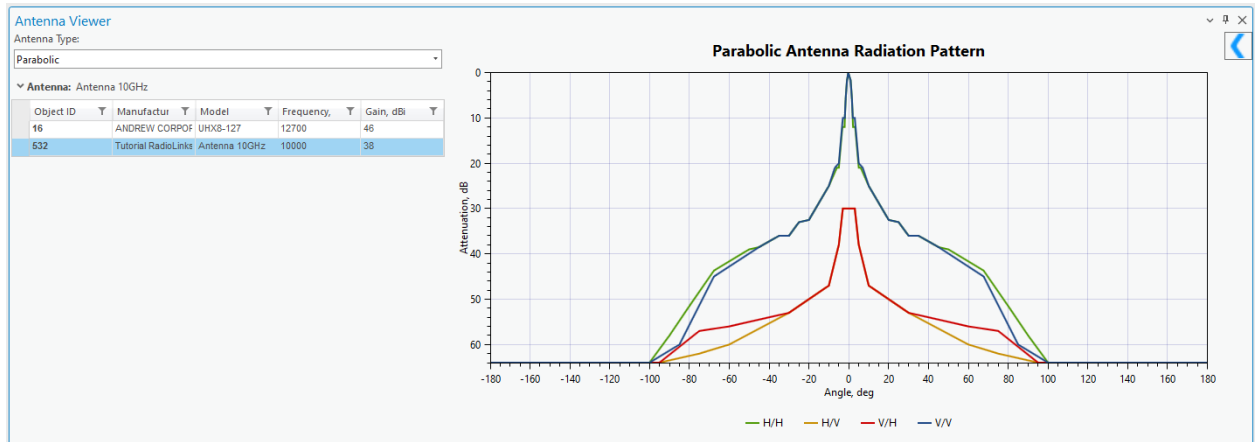


Browse to C:\CE_Course\RL_Prediction\Equipment\Antennas, select a2119.adf antenna and press OK button to add it.



Select Check box near the antenna and press Import Antennas button.

Now open Antenna Viewer tool to review these antennas.

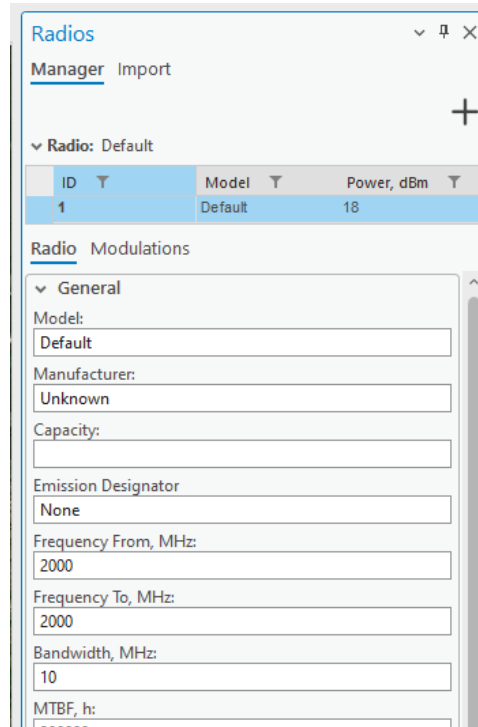


Close Antenna Viewer and Import tools.

3.2 Radios

This equipment category encompasses details about radio transceivers utilized in microwave links. It comprises information on transmitter power, receiver sensitivity, noise figure, nonlinearity characteristics, and maximum data capacity.

Open Radios tool in CE RLP tab and preview available Radio within default CE workspace.



Radios [v] [f] [x]

Manager Import

+

▼ Radio: Default

ID	Model	Power, dBm
1	Default	18

Radio Modulations

▼ General

Model: Default

Manufacturer: Unknown

Capacity:

Emission Designator: None

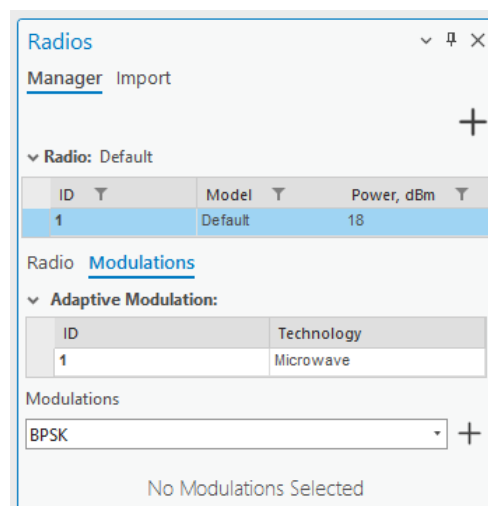
Frequency From, MHz: 2000

Frequency To, MHz: 2000

Bandwidth, MHz: 10

MTBF, h: 100000

It has parameter section, and Modulations section.



Radios [v] [f] [x]

Manager Import

+

▼ Radio: Default

ID	Model	Power, dBm
1	Default	18

Radio Modulations

▼ Adaptive Modulation:

ID	Technology
1	Microwave

Modulations

BPSK +

No Modulations Selected

Click on Add button in top right corner of the dialog.



Define the same parameters as defined below:

- Model: RL Radio
- Manufacture: CE

- Frequency From, MHz: 9500
- Frequency To, MHz: 10500
- Bandwidth, MHz: 10
- MTBF, h: 300000
- MTTR, h: 24
- Bit Rate, kbps: 2048
- Block Size, bits: 2048
- Burst Errors: 1
- Dispersive Fade Margin, dB: 50
- BER 10-3 Threshold, dBm: -76
- BER 10-6 Threshold, dBm: -72
- Noise Figure, dB: 5
- Residual BER: 1E-12
- IIP2, dBm: 30
- IIP3, dBm: 29
- XPIF, dB: 20
- Power, dBm: 20
- Power Low, dBm: 10
- Power High, dBm: 26
- Automatic Transfer Power Control Range, dB: 20

Radios

Manager Import

Radio: Default

ID	Model	Power, dBm
1	Default	18

Radio

Modulations

General

Model:

RL Radio

Manufacturer:

CE

Capacity:

Emission Designator

None

Frequency From, MHz:

9500

Frequency To, MHz:

10500

Bandwidth, MHz:

10

MTBF, h:

300000

MTTR, h:

24

Bit Rate, kbps:

2048

Block Size, bits:

2048

Burst Errors:

1

Dispersive Fade Margin, dB:

50

Receiver

BER 10⁻³ Threshold, dBm

-76

BER 10⁻⁶ Threshold, dBm

-72

Noise Floor, dB:

-103.97722915699808

Noise Figure, dB:

5

Receiver Noise, dBm:

-98.97722915699808

Residual BER:

1E-12

IIP2, dBm:

30

IIP3, dBm:

29

XPIF, dB:

20

Transmitter

Power, dBm:

20

Power Low, dBm:

10

Power High, dBm:

26

Automatic Transfer Power Control Range, dB:

20

Create

Save Changes

Dismiss

Click on Modulations tab, and include these modulations:

Modulations

BPSK

+

> QAM0016

x

> QAM0064

x

> QAM0128

x

> QAM0256

x

You can do it simply selecting modulation from the list and press + button to add it for the radio.

Each modulation has its own default parameters.

QAM0016

Technology: Microwave

Name: QAM0016

Sensitivity, dBm: -73

SNR, dB: 15

Code Rate: 0

Bps, Hz: 0

Downlink Throughput, Mbps: 0

Uplink Throughput, Mbps: 0

Downlink Bitrate, Mbps: 0

Uplink Bitrate, Mbps: 0

Leave default parameters and press Create button. The new radio will be added in the list.

Radios

Manager Import

+

Radio:

ID	Model	Power, dBm
1	Default	18
2	RL Radio	20

Click on Import tab.

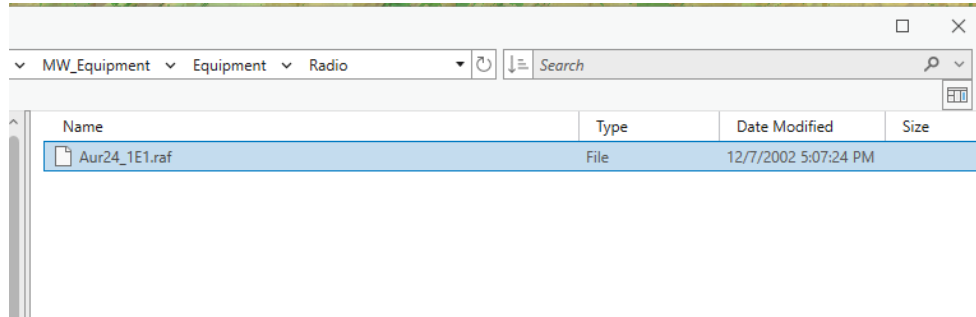
Radios

Manager Import

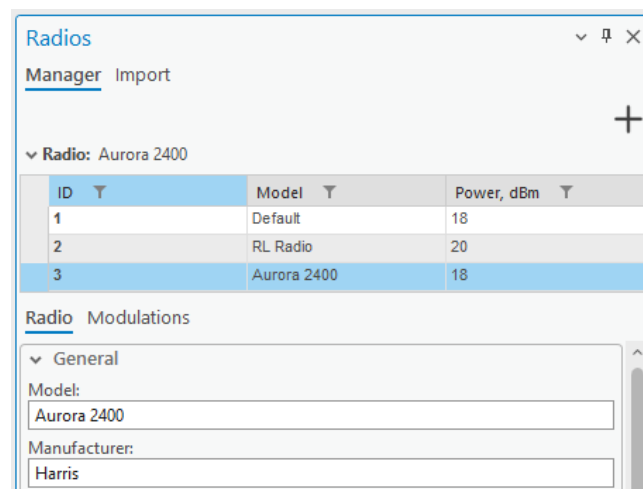
Select Data File (X)

Select Data File to Import

Click on Selected Data File and navigate to C:\CE_Course\MW_Equipment\Equipment\Radio, select Aur24_1E1.raf and press OK button.



New radio with the parameters will be added to the dialog. Press Import button to import it to the database.



Close Radio tool.

3.3 Frequency plans

Frequency planning is important for microwave link design, optimizing spectrum use, and preventing interference. By strategically allocating frequency bands, it enhances efficiency, complies with regulations, and coordinates with other services. This planning considers propagation characteristics and facilitates scalability, ensuring optimal performance for microwave links in various environments and supporting future technology upgrades.

Frequency plans can be imported from a text file, or created manually.

3.3.1 Create manually

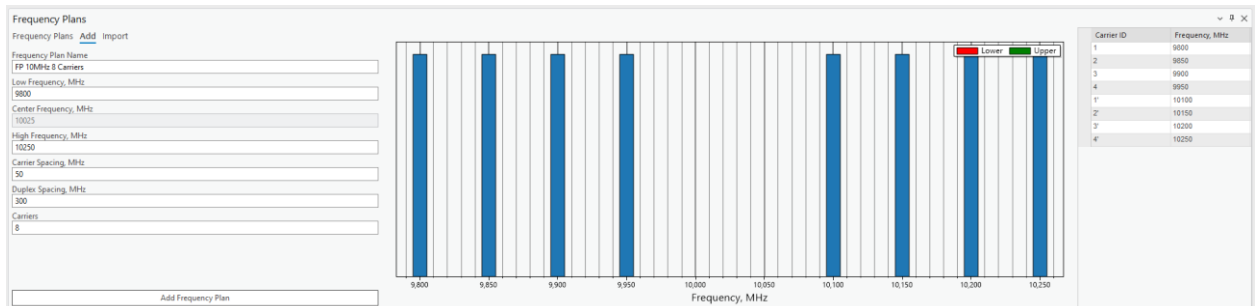
Open Frequency Plans tool in CE RLP tab.



Click on Add section and define:

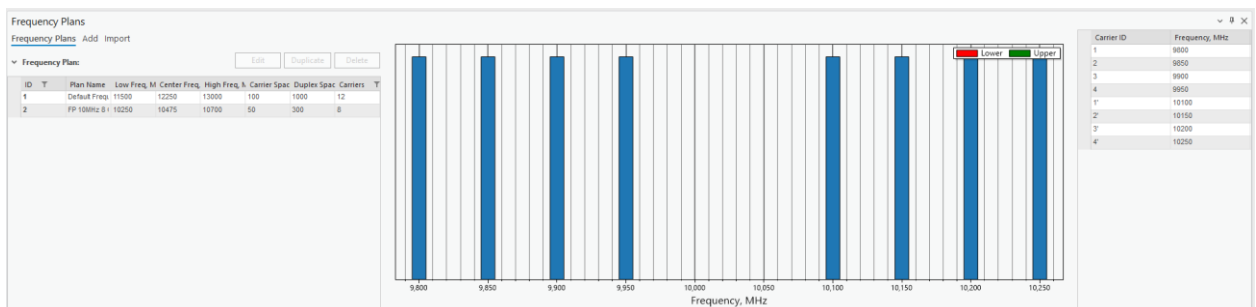
- Frequency Plan Name: FP 10MHz 8 Carriers
- Low Frequency, MHz: 9800
- Carrier Spacing, MHz: 50
- Duplex Spacing, MHz: 300
- Carriers: 8

High Frequency, MHz will be filled automatically.



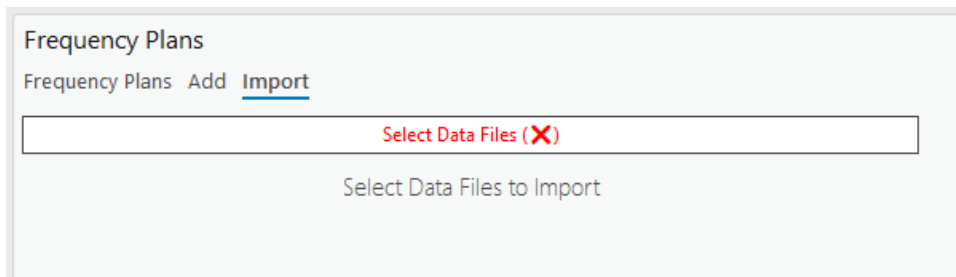
Press Add Frequency Plan button.

The new frequency plan will appear in the main dialog.

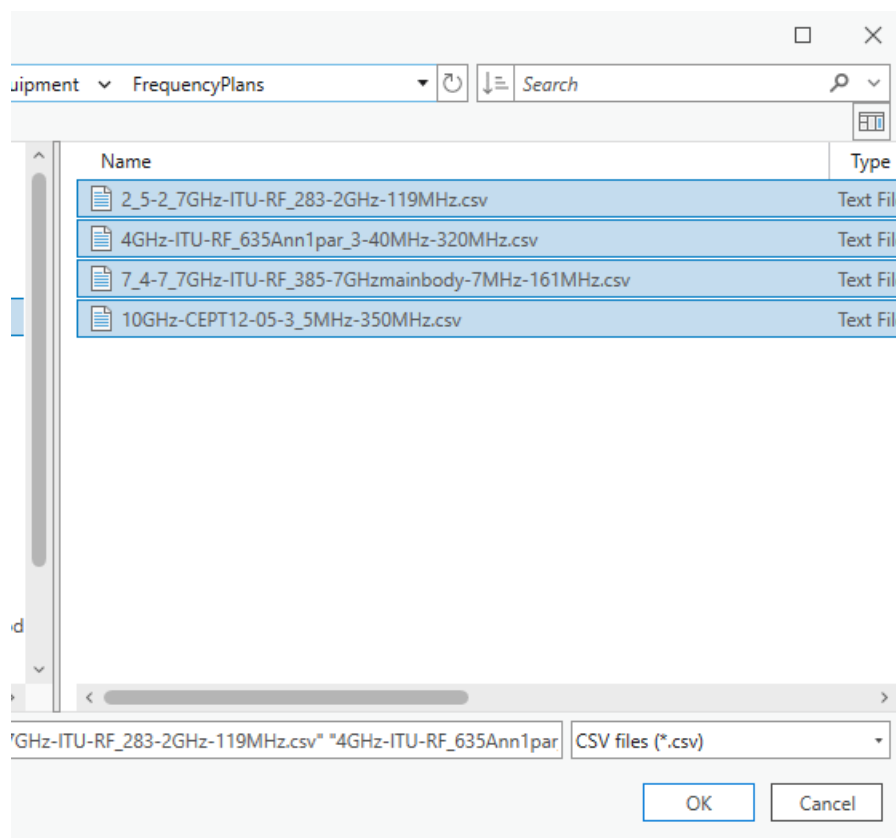


3.3.2 Import

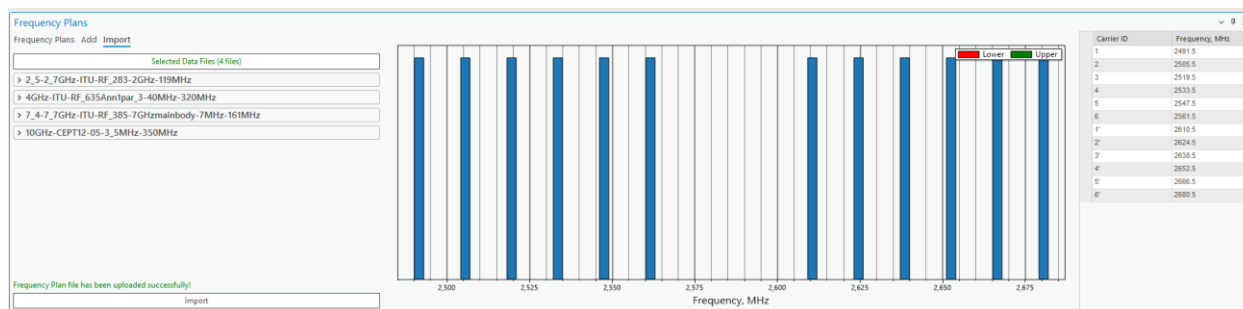
Click on Import option, and then on Select Data Files.



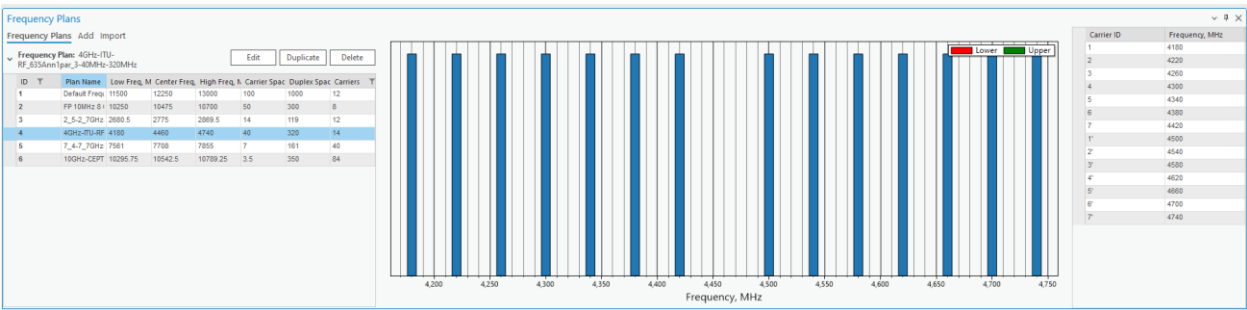
Navigate to C:\CE_Course\RL_Prediction\Equipment\FrequencyPlans, select all files and press OK button.



Frequency Plans will be added to the preview.



Press Import button and they will be imported to the database.



Close Frequency Plans dialog.